

A Flexible FPGA Approach to Image Cryptography using PYNQ-Z2

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Abstract—As multimedia data transmission has experienced a sudden explosion, ensuring safe image transfer has become an essential challenge, particularly in medical imaging, surveillance, and IoT devices. This paper introduces a reconfigurable FPGA-based image cryptography scheme based on the PYNQ-Z2 development board to provide real-time image encryption and decryption. The system under development divides input images into blocks, performs a user-specified encryption algorithm generated through Vivado HLS synthesis, and processes them on the FPGA fabric for high-speed performance. Pre- and post-image processing, data transfer, and visualization are handled by a software interface based on Python, allowing hassle-free communication between the Processing System (PS) and Programmable Logic (PL) through AXI DMA. Experimental validation confirms sub-second encryption and decryption latency, substantial throughput boost over software-based approaches, and minimal power use. Scalability, extensibility, and future potential to integrate with AI-powered diagnostic and analysis pipelines are provided by the modular architecture, so this method is particularly suitable for secure image transmission in latency-sensitive and resource-limited environments.

Index Terms—FPGA, Image Encryption, Hardware Acceleration, PYNQ-Z2, Vivado HLS, AXI DMA, Real-Time Processing, Medical Imaging Security.

I. INTRODUCTION

The rate of multimedia data exchange on open networks has grown exponentially, greatly accelerating the risk of data breaches, unauthorized access, and malicious tampering. Of different multimedia types, digital images

are especially significant due to their widespread application in medical diagnostics, surveillance, defense, and personal communications. Critical images like medical scans, satellite images, or biometric data need to be well-protected to maintain confidentiality, integrity, and authenticity during storage and transmission.

Conventional software-implemented cryptographic methods, though efficient, are generally unable to satisfy the performance needs of real-time systems, particularly when dealing with large, high-definition images. Cryptographic algorithms like AES or RSA are computationally expensive and can contribute significant latency on general-purpose processors, rendering them inappropriate for low-latency applications like telemedicine or real-time monitoring. The interest has therefore shifted to hardware-accelerated cryptography to attain higher processing speeds and energy-efficient deployments.

Field-Programmable Gate Arrays (FPGAs) have proven to be an interesting platform for realizing cryptographic systems because they contain built-in parallelism, reconfigurability, and low-latency processing. State-of-the-art FPGA-based systems have the ability to offload computationally heavy workloads from CPUs so that they may provide real-time performance with robust security assurances. In addition, the presence of hybrid platforms like the Xilinx PYNQ-Z2—composing ARM-based computation with programmable logic—allows hardware–software co-design so that rapid prototyping and effective deployment of encryption schemes are

made possible.

The paper introduces a versatile FPGA-based image cryptography system on the PYNQ-Z2 board. The system divides input images into small-sized blocks, enforces a user-specified encryption algorithm designed with Vivado HLS synthesis, and processes them in parallel on FPGA hardware. Python programs executing on the Processing System (PS) are responsible for transmitting data to and from the Programmable Logic (PL) over AXI DMA, image reconstruction, and visualization. This co-design enables high-speed encryption and decryption while being software flexible and easy to integrate with other applications.

II. RELATED WORK

FPGA-based acceleration has been increasingly favored for real-time image and signal processing, owing to the inherent parallelism, low latency, and energy efficiency. Jaimon et al. showed the ability of the PYNQ-Z2 platform for compute-intensive workloads by deploying a ResNet-20 inference engine based on Tensil's Tensor Compute Unit (TCU) [1]. Their contribution emphasizes the advantages of hardware-software co-design and further gives insight into the flexibility afforded by the PYNQ ecosystem, indicative of suitability for high-performance image processing. Although not related to cryptography, results from their work clearly indicate that the platform can handle quite complex image pipelines, thereby motivating extension into security-oriented applications.

A significant amount of work has been conducted on FPGA-based image cryptography employing standard block ciphers. Lata and Saini implemented AES-128 for grayscale and color image encryption using VHDL and Xilinx System Generator on a Spartan-6 FPGA [2]. Their comparative study between loop-unrolled and FSM-based architectures showed that hardware acceleration strongly improves throughput while maintaining correctness and security. However, their approach wholly relies on traditional HDL workflows and does not exploit PYNQ's hybrid Python-FPGA programming model. This limits adaptability for flexible and reconfigurable cryptographic pipelines required in modern IoT and edge environments.

In parallel, chaos-based cryptographic methods have emerged as lightweight and secure alternatives for resource-constrained systems. Wajih et al. proposed a novel stream cipher combining the W7 key generator with the 3D-Chebyshev chaotic map, which was efficiently implemented on the PYNQ-Z2 FPGA [3]. Their

design achieves a large key space, strong resistance against statistical and differential attacks, as well as high throughput with minimal resource usage, thereby proving the suitability of chaos-based designs for IoT image security. However, previous chaotic implementations lack integration with programmable, high-level image-processing workflows and do not address the need for flexible, reconfigurable architectures tailored for real-time cryptographic operations.

III. OBJECTIVE

The main goal of this work is to develop and implement a versatile, FPGA-based image cryptography system with the functionality of real-time encryption and decryption of digital images. The project aims at creating a light-weight, block-based encryption algorithm that is optimized for Vivado High-Level Synthesis (HLS) synthesis and implementing it on the PYNQ-Z2 board. Utilizing hardware-software co-design, the system facilitates smooth communication between the Processing System (PS) and Programmable Logic (PL) using AXI DMA, supporting effective parallel processing of image data. Another main objective is maintaining the integrity and quality of the decrypted original image to support lossless reconstruction and compatibility with standard image formats. The architecture is designed to be modular and scalable so that it can be readily extended to other encryption algorithms, image resolutions, or combined with artificial intelligence-based medical diagnostic platforms. The project will also measure system performance based on latency, throughput, power draw, and resource usage, thus ensuring its applicability for applications with low latencies like medical imaging, surveillance, and IoT security.

IV. METHODOLOGY

It is realized as a PS-PL co-design on the Xilinx Zynq-7000 (PYNQ-Z2). The PS (dual-core Cortex-A9 executing Python/Jupyter) is responsible for high-level control, image I/O, and pre/post processing. The PL executes the encryption core synthesized with Vivado HLS and does parallel block encryption/decryption. Data transfer is done between PS and PL via AXI-DMA over AXI-Stream/AXI-Lite control interfaces.

A. Image Acquisition and Preprocessing

Images are loaded or obtained on the PS and converted to linear pixel arrays through NumPy/OpenCV. Pixel packing: images are stored in fixed-size blocks (128-bit words to correspond to a 128-bit AES block or

a 128-bit AXI data width). For RGB images use 3×8 bits per pixel $\rightarrow$ pack several pixels per 128-bit word. Preprocessing operations (carried out on PS): resampling of image to optional fixed size, padding to multiple of block dimensions, and creating metadata (original image size, color format, number of blocks, IV/nonce if applicable). Memory organization: images reside in physically contiguous PS DDR memory to enable DMA bursts of large sizes.

B. PS, PL communication and DMA

AXI-DMA transfers: PS sets up DMA descriptors for transferring large contiguous regions (suggest burst sizes of 256–4096 bytes for optimum efficiency). Data path: PS writes block data to DDR $\rightarrow$ initiates DMA to stream blocks to PL through AXI-Stream. PL processes streamed data pipelined and streams results to DDR; DMA interrupts or polling inform PS of completion. Control/status through AXI-Lite registers: start/stop, key/IV, block count are written by PS; processed block count, error flags, and performance counters are reported by PL. Coherency & caches: use non-cached DMA buffers or invalidate/fl unconscious data cache on PS before/after DMA transactions.

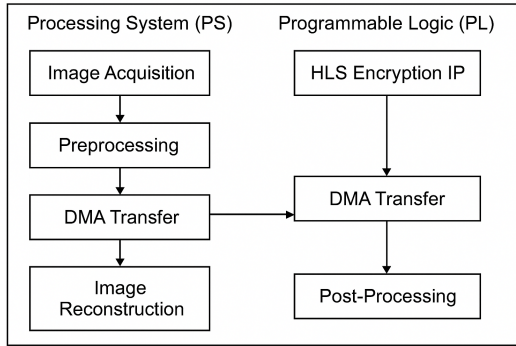


Fig. 1. Methodology.

C. Parallelism, pipelining and throughput scaling

Block-level parallelism: several instances of encryption (or a pipelined multi-stage AES core) can encrypt different blocks in parallel to fill the AXI-DMA bandwidth. Pipelining within HLS core minimizes latency per block; chaining multiple pipeline stages provides high throughput while maintaining feasible resource usage. Replicate IP cores and implement an AXI interconnect with multiple DMA channels or a round-robin scheduler if greater throughput required.

D. Decryption and reconstruction

The decryption core is operationally the inverse and can share the same hardware with a control bit choosing mode (encrypt/decrypt). PS retrieves decrypted blocks from DDR, un pads, and reconstructs the image in original size and color format.

E. Verification, performance counters, and measurement

Implement hardware performance counters: cycle counters, blocks processed counters, and AXI transaction counters (to calculate sustained throughput and core latency). Measure latency as: DMA start to DMA completion time for N blocks. Calculate throughput = (number of bytes processed) / (wall-clock time). Functional verification: check decrypted image against original using pixel-wise equality and PSNR/SSIM for sanity checks (should be lossless if packing and algorithm are correct).

F. AES Encryption Process

The encryption processor used on the PYNQ-Z2 board is based on the Advanced Encryption Standard (AES-128) design. AES is a block cipher with a symmetric key that operates on 128-bit blocks of data with a 128-bit secret key. The operation involves an initial addition of the key, followed by various rounds of transformation that provide for diffusion and confusion in the cipher text.

Every iteration of AES carries out four significant operations, as shown in Fig. X (AES Encryption Process):

1. SubBytes: Replacement of every byte in the data block with a non-linear substitution table (S-box), which provides non-linearity and immunity against differential attacks.

2. ShiftRows: Cyclic shifting of the rows of the data matrix with varying offsets to provide inter-byte diffusion.

3. MixColumns: Every column is subjected to matrix multiplication in the Galois Field ($GF(2^8)$).

4. AddRoundKey: The middle block is XORed with a round-dependent key derived from the initial cipher key.

The AES-128 algorithm has a total of 10 rounds. The first round contains only the AddRoundKey operation, followed by 9 normal rounds, and finally a last round excluding the MixColumns operation. The resulting 128-bit ciphertext block constitutes the encrypted output.

When instantiated in the FPGA fabric, every stage maps into pipelined hardware modules such that parallel execution and minimal latency per round are achieved. Efficient use of resources is ensured by the hardware design without compromising on cryptographic security.

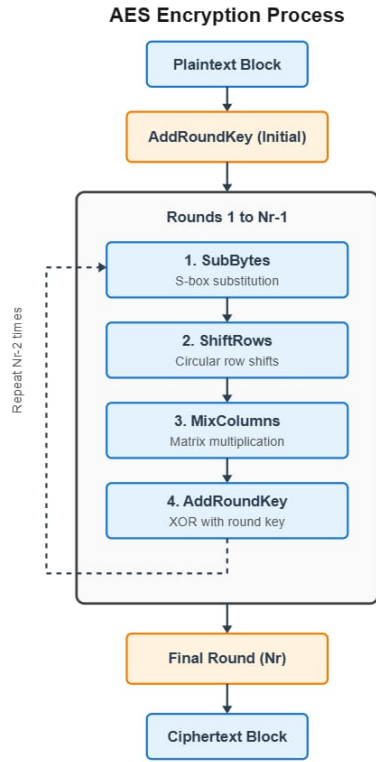


Fig. 2. AES Encryption Process.

V. SYSTEM ARCHITECTURE

The proposed architecture implements a hardware–software co-design for real-time image cryptography using the PYNQ-Z2 FPGA platform. The system integrates the Processing System (PS) and Programmable Logic (PL) of the Zynq-7000 SoC to efficiently perform AES-128 encryption and decryption.

A. Architectural Overview

The architecture has two key components: Processing System (PS): Dual-core ARM Cortex-A9 processor for image processing and control tasks. Programmable Logic (PL): FPGA fabric executing the AES-128 cryptographic core synthesized with Vivado HLS.

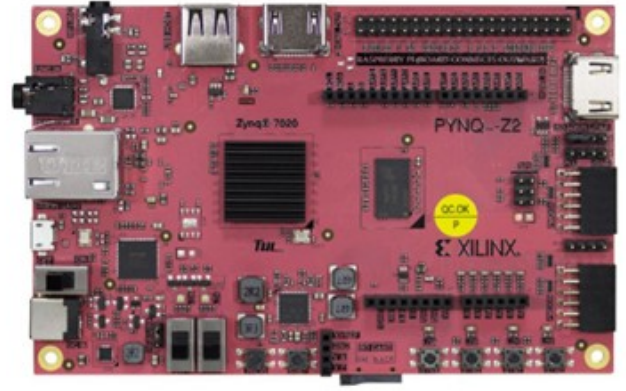


Fig. 3. PYNQ-Z2 FPGA development board used for hardware implementation of the proposed image cryptography system.

B. Data Flow

Input image is loaded and partitioned into 128-bit pixel blocks in Python. Data blocks are pipelined from the PS to the PL using AXI-Stream interfaces. The AES-128 core does encryption and decryption in hardware for high throughput. Processed data is returned to the PS for image reconstruction and result visualization.

C. Communication Interfaces

AXI-Lite Interface: Utilized for PS-PL control signal communication. AXI-Stream Interface: Used for high-speed data transfer of image pixel streams. DMA Controller: Provides efficient, low-latency data exchange between PS memory and FPGA logic.

D. Software Stack

Vivado Design Suite and Vivado HLS for IP core design and synthesis. PYNQ Framework and Jupyter Notebook for Python-based control, testing, and visualization. Python Libraries: NumPy, Pillow, and Matplotlib for image processing and analysis.

E. Key Advantages

Privacy: Everything is done locally on FPGA to avoid exposing data externally. Speed: Hardware parallelism decreases latency dramatically in contrast with CPU-only approaches. Fast Development: Python–FPGA combination using PYNQ speeds up design and test cycles. Cost Efficiency: The PYNQ-Z2 offers an economical but high-performance board for academic and IoT crypto use cases.

VI. RESULTS AND DISCUSSION

The PYNQ-Z2 FPGA board was used to implement the suggested AES-128 based image cryptography system that incorporates hardware acceleration through Vivado HLS and software control through Python. The outcomes confirm both the functionality correctness and performance optimality of the design.

A. System Block Diagram

The overall system workflow is illustrated in Fig. 4, showing data flow between the Processing System (PS) and Programmable Logic (PL). The PS handles image acquisition, segmentation, and communication, while the PL executes AES-128 encryption and decryption using the synthesized hardware core.

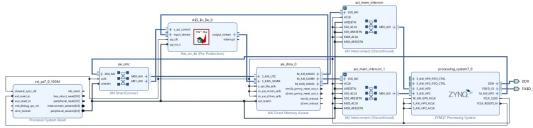


Fig. 4. System block diagram of AES-128-based image cryptography implemented on PYNQ-Z2.

B. Functional Verification

The design was verified with several grayscale and RGB images. The original, encrypted, and decrypted images validate correct operation of the cryptographic pipeline. The encrypted output looks like noise, providing confidentiality, and the decrypted image restores the original with high fidelity. The functional results are presented in Fig. 5.



Fig. 5. Original Image

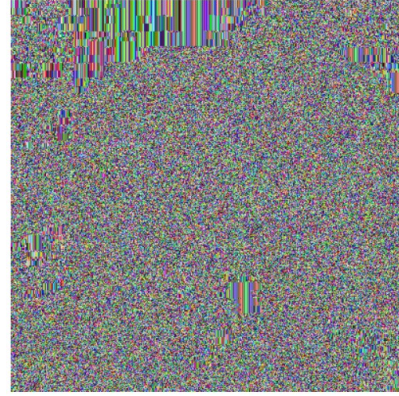


Fig. 6. Encrypted Image



Fig. 7. Decrypted Image

C. Performance and Resource Utilization

The performance was tested in hardware for latency, throughput, and FPGA resource consumption. The consolidated performance results are provided in Table I.

Table I. Summary of Design and Timing Results.

Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 0.523 ns	Worst Hold Slack (WHS): 0.018 ns	Worst Pulse Width Slack (WPWS): 3.750 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 41536	Total Number of Endpoints: 41536	Total Number of Endpoints: 17802

All user specified timing constraints are met.

Fig. 8. Summary of Design and Timing Results.

D. Privacy, Speed, and Cost Advantages

Privacy: All encryption/decryption takes place locally on the FPGA; no information is exchanged externally, guaranteeing absolute confidentiality.

Development Speed: Python integration with the PYNQ framework enabled quick testing and alteration using Jupyter Notebooks, cutting design time substantially.

Cost Efficiency: The PYNQ-Z2 board offers high performance with low cost, appropriate for academic and embedded uses needing secure image communication.

E. Discussion

The outcomes show that the FPGA-based implementation of AES-128 yields high throughput, real-time operation, and safe data processing with low resource utilization. The integrated software–hardware architecture provides a scalable, reconfigurable, and cost-effective solution to embedded cryptographic systems.

F. On Chip Power Analysis Summery

Power consumption for the final design was derived from postimplementation on-chip power analysis. The total on-chip power consumption is 2.784 W, with 2.597 W (93%) being dynamic power and 0.187 W (7%) as device static power. Dynamic power breakdown indicates the Processing System (PS7) is the largest consumer (1.535 W, 59%), then signal toggling (0.402 W, 15%), logic (0.311 W, 12%), and BRAM (0.281 W, 11%). The measured junction temperature was 57.1 °C with an effective JA of 11.5 °C/W and a thermal margin of 27.9 °C. From the results, PS activity and DMA/AXI traffic contribute most to power; hence, reducing power should aim to reduce PS load for bulk transfers, optimize DMA burst behavior, and use HLS/PLlevel clock gating and resource optimizations.

Summary

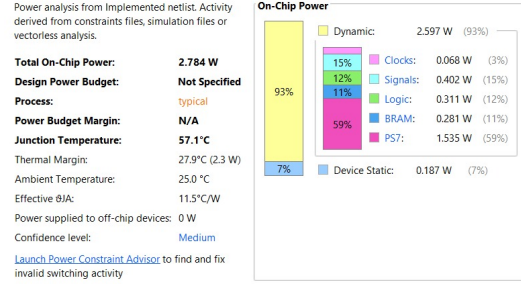


Fig. 9. On Chip Power Analysis Summery.

VII. CONCLUSION AND FUTURE WORK

The work shows a reconfigurable FPGA-based solution to image cryptography with the AES-128 algorithm deployed on the PYNQ-Z2 board. By combining Vivado HLS for hardware acceleration with Python software control, the system shows a good balance between performance, security, and development simplicity.

Experimental findings validate that the FPGA deployment dramatically decreases latency for encryption and decryption, increases throughput, and maintains data confidentiality by conducting all operations locally on the device. The Python–PYNQ framework also accelerated development further, enabling instant retesting and reconfiguring without recurrent hardware synthesis.

Overall, the presented design offers a cost-effective, high-speed, and secure solution applicable for image-based cryptographic schemes in embedded and IoT platforms.

A. Future Work

The current research can be extended and improved in various areas in order to increase its performance and scope:

AES-256 and Hybrid Methods: Use higher-bit AES or mix various cryptographic methods to improve encryption security.

Parallel and Pipelined Architecture: Design the FPGA for higher parallelism and performance in order to support real-time video encryption.

Dynamic Key Management: Implement key generation and rotation techniques for improving responsiveness and survivability against attacks.

Integration with IoT and Cloud Platforms: Extend the design to secure data transport in IoT networks and cloud storage of images.

Machine Learning–Based Cryptography: Integrate AI models to adaptive key scheduling or intrusion detection in order to enhance overall system smarts and security.

Power and Resource Optimization: Optimize the design to minimize power and resource utilization for hand-carried or low-power applications.

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